

# The Conscription Premium: Extensive or Intensive Margin?

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## Abstract

One explanation for the conscription wage premium is that conscripts accumulate valuable human capital during their service. An alternative explanation is that the premium operates through the extensive margin, having served versus not, rather than the intensive margin of how long service lasts. This paper distinguishes between these mechanisms by estimating the effect of military service duration on income, employment, and educational attainment using data from South Korea and an instrumental variables approach. By restricting the sample to individuals who served, I hold the extensive margin fixed and isolate variation in service duration to identify intensive margin effects. I confirm a conscription wage premium: veterans earn more than non veterans, particularly among less educated men. However, service duration has no effect on income or employment, while longer service reduces the likelihood of obtaining a postgraduate degree, consistent with human capital depreciation rather than accumulation. These findings suggest that the conscription wage premium operates primarily through the extensive margin rather than through human capital gained from longer service.

# 1 Introduction

A large body of literature, such as [Lim \(2018\)](#) and [Lee and Jung \(2020\)](#), studies the effect of conscription on labor market outcomes and educational attainment. They find conscription positively affects those outcomes, but few examine the mechanism behind such an effect. Similar to a college degree, on the intensive margin, finishing compulsory military service (CMS) can help conscripts accumulate valuable human capital. On the extensive margin, conscription can be a signaling or networking device, it can also confer a sense of belonging among other functions. There seems to be no consensus on the nature of higher education ([Huntington-Klein, 2021](#)). In this paper, I study which mechanism is mainly behind conscription's effect on a veteran's labor market outcomes and educational attainment. To study the issue, I use an IV approach and a rich dataset from South Korea to determine if the duration of CMS affects an individual's income, employment, and educational attainment. I identify the causal impact of conscription length on those outcomes by exploiting the exogenous variation in the duration due to North Korean provocation events. These events increase the duration of service but are plausibly unrelated to other unobserved determinants of outcomes. By focusing only on those who served as a conscript and examining the impact of the duration of CMS, I am able to control for the singling effect and study only the channel affecting human capital accumulation.

Many countries such as Russia and South Korea still rely on conscription for military service. Understanding the effect of conscription on the labor market and education outcomes is still important because it will help those countries form relevant policies. Existing literature such as [Bauer et al. \(2012\)](#) and [Bingley et al. \(2022\)](#) disagreed on whether conscription offers a wage premium to veterans. The paucity of literature examining the mechanism behind conscription makes it even more difficult to determine if conscription does come with a wage premium. More importantly, taking a break during work or study to serve as a conscript can be detrimental to one's labor market outcomes and educational attainment. If little or no positive human capital

accumulation occurs during conscription, governments extending the length of conscription will be even more harmful for marginalized men.

As mentioned earlier, conscription may lead to human capital accumulation on the intensive margin. It may also affect labor market outcomes on the extensive margin. If conscription primarily affects labor market outcomes through the extensive margin, then the duration of service, which impacts only the intensive margin, should not positively influence those outcomes. As a country with one of the longest duration of CMS, South Korea has frequently changed the lengths of its conscription over the years. However, different male cohorts with different duration of service may face unique economic shocks that are correlated with individuals' labor market and education outcomes, which can be a potential source of bias. To examine how the length of the conscription in South Korea affects conscripts' income, education attainment, and employment status, I use the moving averages of North Korean (NK) provocation events when conscription started for each cohort as the instrument variables and the individual-level data from the Korean Labor and Income Panel Study (KLIPS).

I start by replicating the results in [Lim \(2018\)](#). Similar to their findings, I find being a conscript is associated with a higher income and this positive effect can only be observed in males with a low level of education. I then use the IV approach to examine the impact of the duration of CMS. I find that the length of conscription appears to have no effect on respondents' income or employment status. However, a longer conscription length does lower male conscripts' chances of acquiring graduate degrees, suggesting human capital depreciation. The evidence suggests that conscription likely affects labor market outcomes on the extensive margin.

## 2 Background

### 2.1 Military Service in South Korea

The South Korean Military Service Law stipulates that all men between the ages of 18 and 38, except those who have a serious physical or mental injury, such as a torn anterior cruciate ligament (ACL), a missing index finger or a serious mental illness, must serve a mandatory military duty term. Exemptions are extremely hard to acquire for healthy males. Only athletes who won international awards and artists who won exceptional awards can obtain an exemption. Healthy males must serve the full length and cannot quit early ([Toyryla, 2023](#)).

An enlisted man can choose a sector of service in the military. He can select among the Army, the Navy (including the Marine Corps), and the Air Force. Once he chooses the branch of the military, he can also choose what rank to apply to: enlisted soldier, non-commissioned officer, or commissioned officer. If he makes no choice, he must perform his service as an enlisted soldier in the Army. All the other options to perform his service must be made during the registration process, and are subject to qualification tests, which are different for each option ([Hofverberg, 2022](#)).

The duration of service also varies slightly between different branches over the years by about 2 months. The duration of the CMS in all branches changes in the same direction and by roughly the same length at the same time. Generally, the service length in the Army and Marine Corps is the shortest. The duration of conscription in the Navy and the Air Force is slightly longer. In some years, conscripts serve a longer term in the Air Force ([Ari, 2019](#)).

The active period of military service has generally varied in South Korea, depending on the geopolitical situation on the peninsula, ranging from 18 to 39 months. North Korean provocations play a major role in determining South Korea's duration of conscription. For instance, the South Korean government extended the duration of conscription for all branches in 1968 when a group of North Korean commandos infiltrated the South and attempted to assassinate the South Korean president ([Ari, 2019](#)).

Lim (2018) compares the wages of male conscripts to those who have never served in South Korea with an IV approach. They find conscripts enjoy a wage premium and it is the less educated men who receive such wage premium. They hypothesize that the wage premium can be from either more human capital accumulation or a positive signal to employers that veterans are potentially good workers. The findings of Lee and Jung (2020) further confirm the existence of the conscription premium in South Korea.

## 2.2 Analytical Framework

Conscription can affect an individual’s income through both the intensive and the extensive margins. On the intensive margin, conscription can lead to human capital accumulation. Bingley et al. (2022) find compulsory military service can lead to skill acquisition in Denmark. They find male conscripts’ numeracy and literacy have significantly improved after conscription. The positive effect of conscription is the strongest for men with low scores in the Armed Forces Qualification Tests.

Savcic et al. (2023) find male conscripts who have served a longer term immediately after high school obtain a higher GPA in university. They believe conscripts acquire non-cognitive skills when they serve a longer term, which improves academic performance. Non-cognitive skills like responsibility, self-discipline, teamwork, perseverance, leadership, self-assurance, and other social skills can also improve one’s labor outcomes such as income. As Lim (2018) points out, South Korean conscripts may also obtain work-specific skills. And longer service length makes it possible for conscripts to hone those skills. Lastly, there is also human capital depreciation in skills such as math and biology when conscripts take a break from work or study.<sup>1</sup>

On the extensive margin, conscription can serve multiple functions. It can send a positive signal about the quality of a worker to potential employers. It may be a networking device that offers better career opportunities. Those who served in the military can also share a common sense of belonging so that they

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<sup>1</sup>Hjalmarsson and Lindquist (2019) find conscripts are more prone to committing crimes suggesting a loss in human capital.

favor one another in the civilian workplace. Lastly, conscription can filter out those who are not mentally and physically healthy.

In all, conscripts can acquire experience or technical know-how during their service through the intensive margin. However, there may also be human capital depreciation. On the extensive margin, conscription can be a signaling or networking device, it can also confer a sense of belonging. To determine through which margin conscription affects labor market outcomes, we can examine how the length of conscription affects labor market outcomes and educational attainment. Conscription length can only affect labor market outcomes and educational attainment through human capital accumulation: the intensive margin. And if the conscription wage premium mainly comes from the extensive margin of conscription, the duration of conscription should have no positive effects on those outcomes.<sup>2</sup>

## 3 Methodology

### 3.1 Data

The bulk of the panel dataset used in the analysis comes from the Korean Labor & Income Panel Study (KLIPS). The individual-level dataset contains detailed information about a respondent’s date of birth, income, education, region of residence, employment status, occupation, age, military service record, and marriage status from 1998 to 2019. The survey relies on questionnaires filled out by 5,000 urban households in cities nationwide (city sub-districts, towns, and sub-counties) in South Korea. Only those who reside in the Jeju province and those who are serving in the military or institutionalized are excluded from the survey. To overcome the representation problems due to sample attrition, an additional sampling of 1,415 households was conducted in 2009. In 2018, an additional 5,044 households were sampled (KLI, 2022).

For data on North Korean provocations, I use data from the Center for

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<sup>2</sup>Puhani and Sterrenberg (2021) find no significant impact of the service length on labor market outcomes.

Strategic and International Studies. It includes missile launches, nuclear tests, incursions, infiltration, hijacking, and other provocations. The center keeps track of all provocation events initiated by North Korea regardless of the number of casualties since 1958 (CSIS, 2019). As will be shown later, physical attacks such as incursions and hijacking primarily affect South Korea’s conscription length compared to missile and nuclear tests<sup>3</sup>. The South Korean military expenditure to GDP ratio is obtained from the World Bank (WB, 2022).

### 3.2 Empirical Strategy

Since only men are subject to mandatory military service and this paper examines the effect of service length, only males are included. I keep only people aged 59 or less because the legal retirement age is 60 years and the decision to retire is endogenous. According to the South Korean Defence White Papers, the military service length ranges from 18 to 39 months over the years. When the South Korean government decides to change the duration of the CMS, they change it for all branches of the military by roughly the same length (Ari, 2019). I keep only respondents whose service length is between 18 and 39 months to minimize the impact of measurement errors. Furthermore, anyone who served before the age of 18 or after the age of 38 is also dropped to minimize measurement errors. Professional military personnel who work in the defense sector are also dropped from the samples.

The duration of conscription varies primarily at the cohort level, and a cohort may be subject to the same socioeconomic shocks that are correlated with the length of conscription and the outcome variables. For example, the government may decide to shorten the duration of the service to deal with a labor shortage. And a labor shortage may drive up the income. So not controlling for such shocks may introduce a downward bias. Another confounding factor may come from the universities. Typically, South Korean men first graduate high

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<sup>3</sup>Indeed, according to Lee (2022), a 2022 survey of 1,006 adults by the Korea Society Opinion Institute (KSOI) and the Kukmin Ilbo newspaper said the North Korean missile launches over the previous month would not influence their choice for the country’s next president.

school, complete one or two years of university studies, and then start their military service (Toyryla, 2023). As service length is shortened, universities may decide to cover more materials in the first two years of universities due to less depreciation of human capital. And more materials covered may improve students' chances of better job placement, which may also bias the results.

The duration also varies slightly at the individual level due to conscripts serving in different branches. It is plausible that some individual characteristics may be correlated with the duration of the CMS. On the other hand, serving in the Air Force or the Navy can mean better living conditions and more specialized knowledge that can potentially raise conscripts' wages. The data set does not have information on the branch in which a respondent served.

To deal with the potential endogeneity, I use the 6-year, 5-year, and 4-year moving averages of all provocation events by North Korea before conscription started for each individual as the instruments. The idea is that a higher level of threat from North Korea may force the South to extend its conscription length to increase its military preparedness.

The two neighboring countries share one of the most heavily defended borders in the world. Since the Korean War, the two countries have had little economic interaction. Wallace (2016) concludes that North Korea engages in provocations for internal reasons. The list of reasons for provocations ranges from domestic economic pressure to garnering domestic support during regime succession. Provocations by the authoritarian regime ultimately maximize the interest of the Kim dynasty. As long as a full-scale war does not break out, provocations from North Korea should be exogenous to other cohort-level and individual confounding factors in South Korea due to the long-term division between the two countries.

Yoo and Kim (2017) find that provocations are mostly driven by internal causes rather than external stimuli, and that North Korean leaders are more likely to utilize provocations in order to demonstrate power when they are politically strong and may have an incentive to divert internal economic discontent when the country is economically weak. They also find South Korean socioeconomic factors such as the presidential election and the ruling party of



the country are not related to the North Korean provocation. North Korea may also attack when they perceive its relative military strength to be strong. By controlling for the defense budget to GDP ratio in South Korea, I can also control for this potential bias.

As shown below, equation 1 represents the second stage.

$$Y_{ict} = \alpha_0 + \alpha_1 \text{Conscription\_Length}_{ic} + \gamma X_{ict} + \text{Region}_j + \text{Year}_t + \epsilon_{ict} \quad (1)$$

Subscripts  $i$ ,  $c$ ,  $j$ , and  $t$  refer to the individual, the year when conscription started, the region, and the year of the survey, respectively. The instruments are the 6-year, 5-year, or 4-year moving averages (MA) of North Korean provocation events before conscription started for each individual.  $\text{Conscription\_Length}_{ic}$  is the length of conscription for each individual in months.  $Y_{ict}$  is the dependent variable. It can be the log inflation-adjusted pre-tax income, whether a respondent holds a post-secondary or post-graduate degree, and whether a respondent is working.  $\text{Region}_j$  and  $\text{Year}_t$  are the region and year fixed effects, respectively.  $X_{ict}$  include the education level, age, the age when conscription started and log defense to GDP ratio when conscription started, the class, and the type of the job of a respondent.  $\epsilon_{ict}$  is the error term.

The error terms for those in the same conscription cohort may be correlated. Similarly, the error terms in every survey year may also be correlated. Because of that, two-way clustering by survey year and year when conscription started (cohort) is also implemented. Clustering at the cohort level means there is no need to cluster at the lower individual level. Since the survey only has information on respondents' province of residence and South Korea has only 16 provinces, there are too few clusters for clustering by region.

Service length is generated using the dates when conscription started and ended. The age when conscription started is generated with the date of birth and the beginning date of conscription. Log military spending to GDP ratio in South Korea is matched with every cohort as control. A cohort is a group of males who started conscription in the same year. Log income is obtained by taking the log of the inflation-adjusted income of each individual. There

are two measures of educational attainment: whether a respondent holds a post-secondary degree or above and whether a respondent holds a post-graduate degree. These dummy variables are equal to one if a respondent holds the respective degree.

Figure 1 shows the first stage results of different moving averages of all North Korean provocation events in the previous years. Starting with the 4-year moving average, North Korean provocations have a positive and significant effect on conscription length. Regardless of the order of the moving average, provocations mostly have a positive and significant effect on service length.

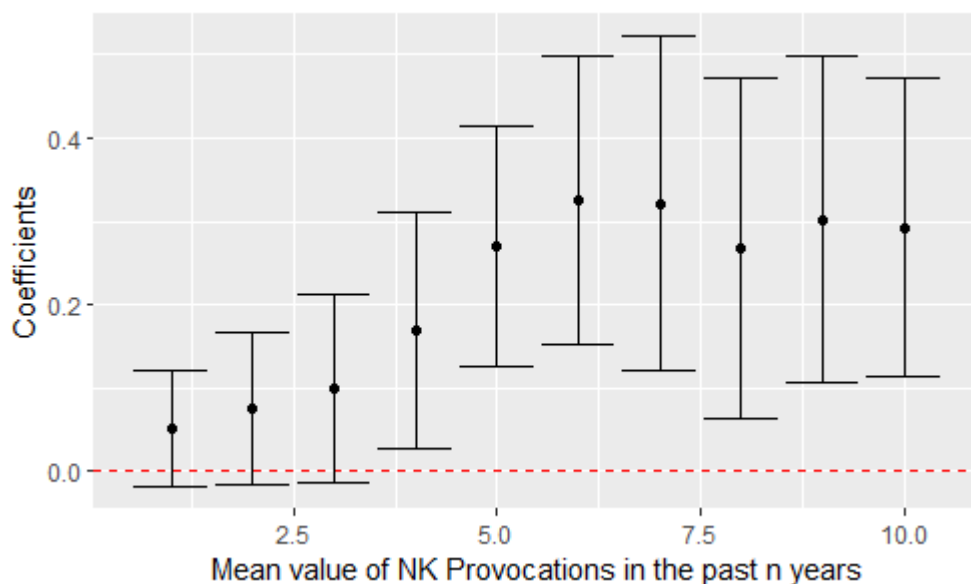


Figure 1: Effects of NK Provocations On SK Conscription Lengths (1)

Next, I include only incursions, physical attacks, and other events that led to actual damages in provocations. Figure 2 (a) contains the first-stage results of this measure of North Korean provocations. The new measure here appears to have a more significant effect than before. It is clear that more provocations from the North in prior years will lead to longer conscription periods even with a different measure of provocations.

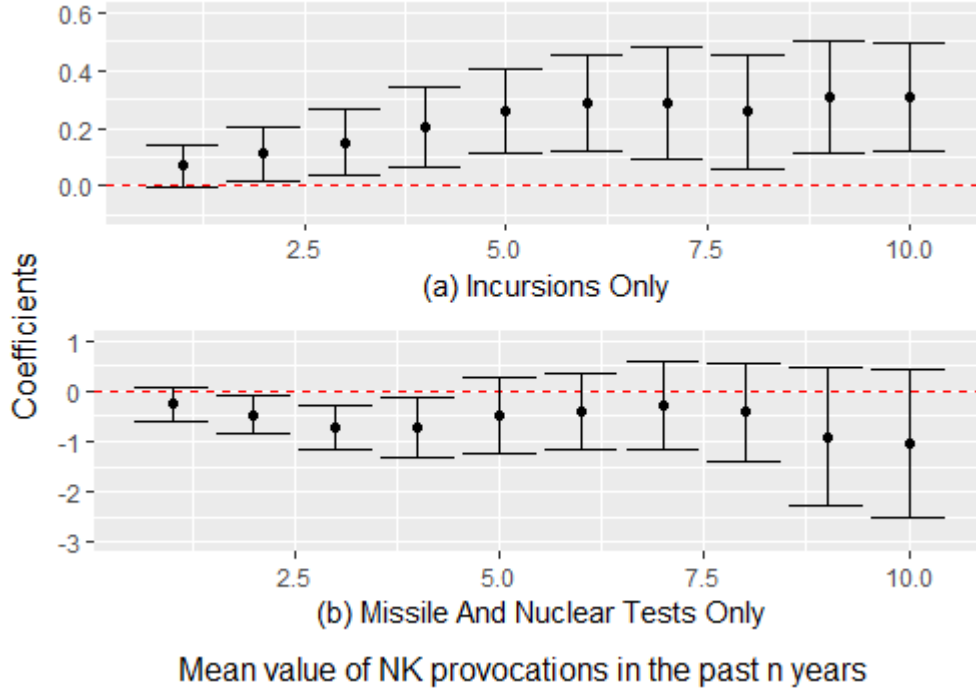


Figure 2: Effects of NK Provocations On SK Conscription Lengths (2)

Lastly, I include only nuclear and missile tests for provocation events. See Figure 2 (b) for the first-stage results of this measure of provocation events. As Lee (2022) reported, South Koreans do not seem too concerned about those missile and nuclear tests. The mostly insignificant effects here corroborate that claim. However, if there are attacks that lead to actual damages, then the situation may escalate.

Figure 3 shows the distribution of the length of the conscription in the sample. Table 1 contains the mean of all important dependent and independent variables by the median length of the service. Conscription length is measured in months. The older generation tends to serve a longer term in the military. As reported by Toyryla (2023), South Korean males finish their compulsory military service during university. The respondents in the sample also started their service around 21 years old, one or two years after starting university around age 19. A longer service length also comes with more military spending. Indeed, higher military preparedness also requires higher military expenditures.

The younger generation who serves a shorter military service is also more educated holding more post-secondary and post-graduate degrees. As shown in Figure 4, the length of the conscription appears to have little effect on income.

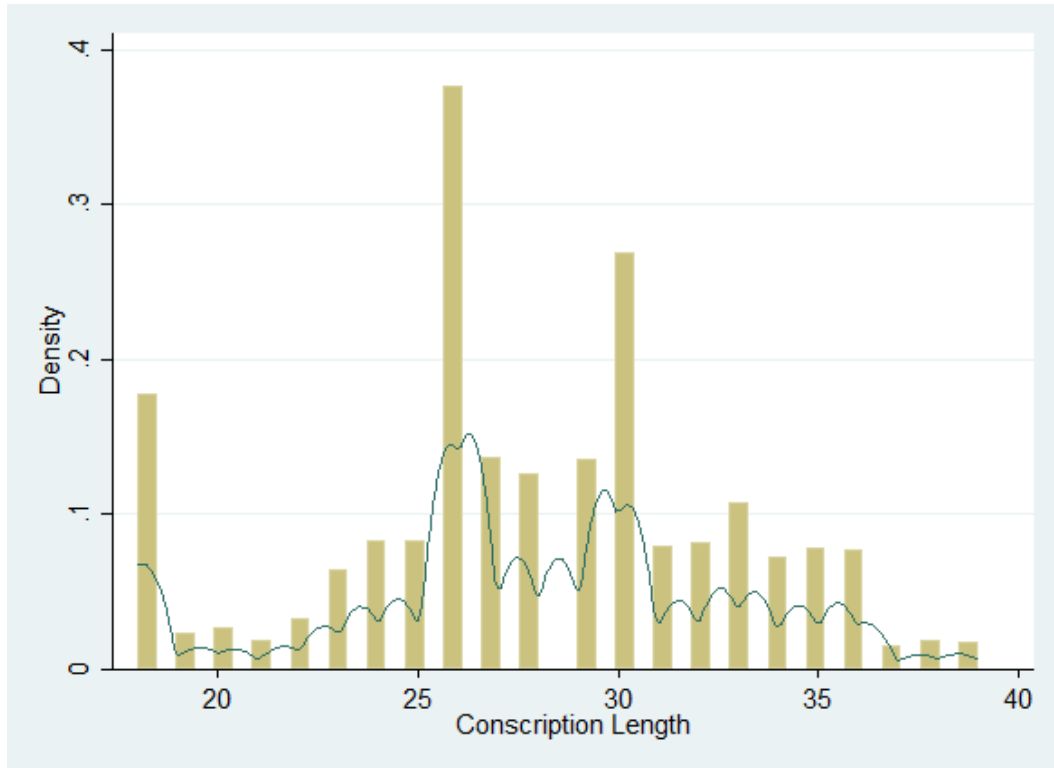


Figure 3: Distribution of Service length

Table 1: Summary Statistics

| (1)<br>Variables                                         | (2)<br>Full<br>sample | (3)<br>Short service<br>length<br>( $< 28$ mos) | (4)<br>Long service<br>length<br>( $> 28$ mos) | (5)<br>P-value of<br>mean-comparison<br>test (col 3&4) |
|----------------------------------------------------------|-----------------------|-------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Age                                                      | 43.87                 | 41.22                                           | 47.06                                          | 0                                                      |
| Age conscripted                                          | 20.78                 | 20.82                                           | 20.74                                          | 0                                                      |
| Length of<br>conscription<br>(months)                    | 27.79                 | 24.24                                           | 32.08                                          | 0                                                      |
| Military spending<br>as a percentage<br>of GDP (%)       | 4.20                  | 3.75                                            | 4.74                                           | 0                                                      |
| Inflation-adjusted<br>pre-tax income<br>(krw 10,000)     | 4,275                 | 4,209                                           | 4,354                                          | 0                                                      |
| Post-secondary<br>Or Post-graduate<br>degree holders (%) | 55.30                 | 65.03                                           | 43.57                                          | 0                                                      |
| Post-graduate<br>degree holders (%)                      | 6.2                   | 7.9                                             | 4                                              | 0                                                      |
| Unemployment<br>rate (%)                                 | 3.49                  | 3.42                                            | 3.56                                           | 0.512                                                  |
| Obs                                                      | 31,231                | 17,081                                          | 14,151                                         | -                                                      |

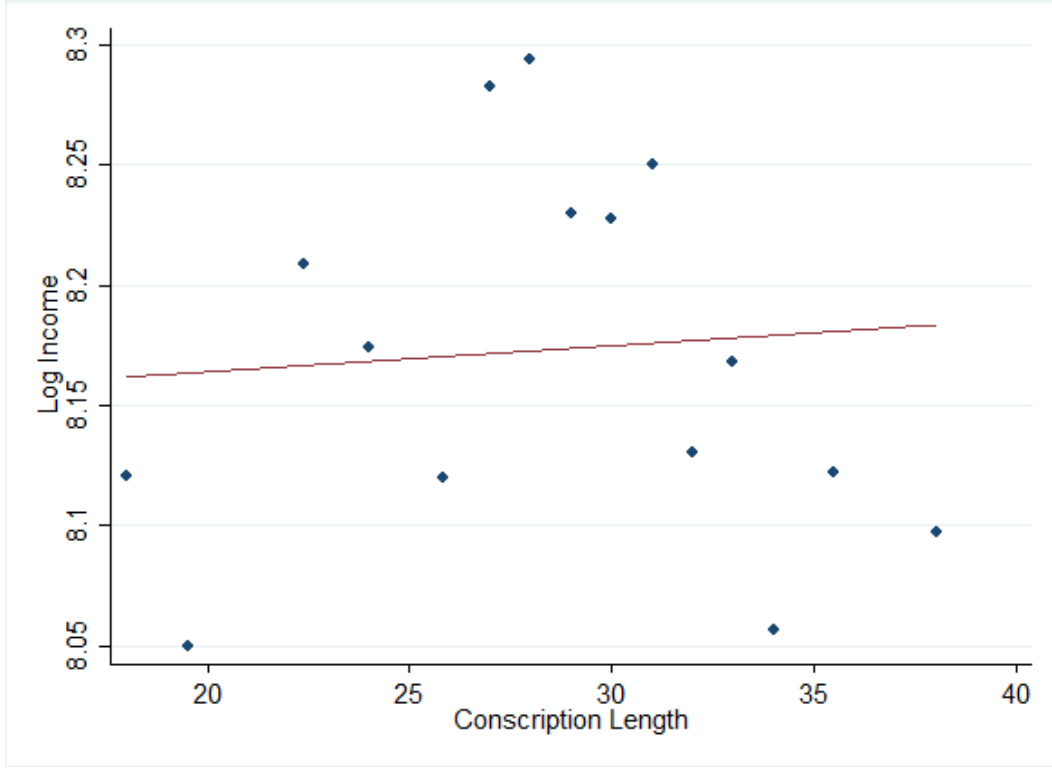


Figure 4: Conscription Length and Income

Note: the graph is a bin scatter plot of log inflation-adjusted income and the duration of conscription generated with the sample.

## 4 Results

### 4.1 Main Results

I start by replicating the results in [Lim \(2018\)](#). The paper compares the labor market outcomes between males who have served in the military and those who never have with OLS and IV models. I regress log income on whether the respondent served in the military and other control variables using an OLS model. All respondents in the sample are males who have at least a high school education. Table 2 has those results. As shown in column 1 of the table, being a veteran is associated with a 4.22% increase in annual income. Columns 2, 3,

and 4 include only respondents with the respective degrees. Only column 2 which includes only high school graduates or lower has a positive and significant result. Similar to what [Lim \(2018\)](#) finds in their OLS and IV models, the wage premium can only be observed for men with a low education level.

Table 2: Replication of [Lim \(2018\)](#): Military Service and Income

| Dependent variable: Log inflation-adjusted income |                     |                                |                                    |                                   |
|---------------------------------------------------|---------------------|--------------------------------|------------------------------------|-----------------------------------|
| Variables                                         | (1)<br>Full sample  | (2)<br>High school<br>or lower | (3)<br>Post<br>secondary<br>degree | (4)<br>Post<br>graduate<br>degree |
| Military service                                  | 0.042***<br>(0.011) | 0.042***<br>(0.016)            | 0.019<br>(0.015)                   | 0.005<br>(0.038)                  |
| Observations                                      | 62,495              | 30,511                         | 28,254                             | 3,730                             |
| R-squared                                         | 0.451               | 0.436                          | 0.435                              | 0.593                             |

Note: The full sample only includes males who have at least a high school education. Military service is equal to 1 if a respondent served as a conscript and 0 otherwise. All models controlled for education, age, the age square, health condition, family economic status, industry, and job type. Year and region fixed effects were also included. The standard errors are clustered at the individual level.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 3 examines the effect of conscription length on income. The instruments are the 6-year, 5-year, and 4-year moving averages of North Korean provocations prior to conscription. Service length is measured in months. The first stage results can be seen in Table 10. It is clear that there are no significant effects in both the OLS and IV models. I find service length has no effect on income.

There are a few potential identification concerns here. The regression may lack power. Factors such as marriage status and the industry respondents work in may be relevant omitted variables. To address those concerns, I will use different specifications and samples in the following robustness checks.

Table 3: Conscription Lengths and Income

| Dependent variable: Log inflation-adjusted income |                  |                  |                  |                   |
|---------------------------------------------------|------------------|------------------|------------------|-------------------|
| Variables                                         | OLS              | IV               |                  |                   |
|                                                   | (1)              | (2)              | (3)              | (4)               |
|                                                   |                  | 6-year MA        | 5-year MA        | 4-year MA         |
| Service length                                    | 0.003<br>(0.002) | 0.018<br>(0.029) | 0.006<br>(0.026) | -0.007<br>(0.035) |
| Weak identification test                          |                  | 12.05            | 11.29            | 4.964             |
| Observations                                      | 31,230           | 31,226           | 31,230           | 31,230            |
| R-squared                                         | 0.309            | 0.299            | 0.310            | 0.306             |

Note: The instruments are the 6-year, 5-year, and 4-year moving averages (MA) of North Korean provocations prior to conscription. Service length is measured in months. All models controlled for education, age, the age at conscription, military spending for each cohort, class, and type of a respondent's job. Year and region fixed effects were also included. The standard errors are clustered by the survey years and the years when conscription started. See the first stage results in Table 10.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## 4.2 Robustness Checks

Although there are multiple observations for each individual, the variable of interest only exhibits cross-sectional variation. Repeated measures for the same individuals can bias the standard errors. By collapsing the data to one observation per individual, I can more accurately reflect the cross-sectional variation. Therefore, I consolidate the data by the individual, retaining only age, age at conscription, defense budget to GDP ratio, and the first non-missing value of education level. The education level is retained because it barely varies. As shown in Table 4, both the OLS and the IV models find no significant effects. The results here also confirm that service length has no effects on income. The first stage results can be found in Table 11.



Table 4: Mean Income and the Duration of Conscription

| Dependent variable: Log mean inflation-adjusted income |                   |                   |                   |                   |
|--------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|
| Variables                                              | OLS               | IV                |                   |                   |
|                                                        | (1)               | (2)               | (3)               | (4)               |
|                                                        |                   | 6-year MA         | 5-year MA         | 4-year MA         |
| Service length                                         | -0.001<br>(0.002) | -0.019<br>(0.014) | -0.017<br>(0.015) | -0.019<br>(0.017) |
| Weak Identification Test                               |                   | 39.55             | 36.90             | 25.95             |
| Observations                                           | 4,300             | 4,297             | 4,300             | 4,300             |
| R-squared                                              | 0.155             | 0.133             | 0.136             | 0.131             |

Note: The dependent variable is log of mean inflation-adjusted income. The instruments are the 6-year, 5-year, and 4-year moving averages (MA) of North Korean provocations prior to conscription. Service length is measured in months. All models control for education, age, the age at conscription, and military spending for each cohort. The standard errors are clustered by the years when conscription started. See the first stage results in Table 11. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Next, I trim off only the top and bottom 5% of the sample by service length. Dropping those observations is necessary because lengths such as 6 and 50 months have never existed after the war according to Ari (2019). The sample now contains respondents with conscription lengths between 14 and 39 months. Fewer control variables are included and the errors are clustered at the cohort level only. As shown in Table 5, the results from all models contain no significant effects.

Table 5: Parsimonious Models

| Dependent variable: Log inflation-adjusted income |                  |                  |                  |                   |
|---------------------------------------------------|------------------|------------------|------------------|-------------------|
| Variables                                         | OLS              | IV               |                  |                   |
|                                                   | (1)              | (2)              | (3)              | (4)               |
|                                                   |                  | 6-year MA        | 5-year MA        | 4-year MA         |
| Service length                                    | 0.001<br>(0.001) | 0.008<br>(0.024) | 0.001<br>(0.029) | -0.020<br>(0.041) |
| Weak identification test                          |                  | 15.59            | 13.95            | 7.547             |
| Observations                                      | 32,734           | 32,729           | 32,733           | 32,734            |
| R-squared                                         | 0.337            | 0.334            | 0.337            | 0.302             |

Note: The instruments are the 6-year, 5-year, and 4-year moving averages (MA) of North Korean provocations prior to conscription. Service length is measured in months. All models controlled for education, age, the age at conscription, military spending for each cohort, and type of a respondent's job. Individual, year, and region fixed effects were also included. The standard errors are clustered by the years when conscription started. See the first stage results in Table 12. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

The results appear to be robust to different specifications. Since everyone in the dataset served in the military, the effects on the extensive margin has been controlled for. Any significant effects of conscription lengths on labor market outcomes should be mainly a result of changes in human capital accumulation on the intensive margin. And no significant effects are observed here. The lengths of conscription do not have an effect on income. And that means conscription affects labor market outcomes primarily through the extensive margin.

### 4.3 The Heterogeneous Effects of Conscription

Lim (2018) finds that the conscription wage premium can only be observed in respondents holding a high school degree or lower. They explain the phenomenon as conscription helping low-skilled workers accumulate human capital that workers can later use in their jobs. To examine if that claim is true, I split the sample by whether respondents hold at least a high school degree. The results of the analysis can be found in Table 6 and the first stage results are

in Table 13. Only the 5-year moving average is used as the instrument for its high weak identification test value.

As shown in Table 6, both the OLS and the IV models find insignificant results for both post-secondary degree holders or above and high school diploma holders or below. The insignificant results reject the claim that low-skilled workers gain much experience during conscription. The wage premiums enjoyed by low-skilled workers are mainly a result of participation in military service instead of serving a longer term. The hardships that conscripts have to endure during military service and the good health that is needed to be in the service are certainly what employers of manual labor are looking for. However, the insignificant results indicate the possibility that there can be a greater impact for individuals with a lower education.

Table 6: Results by Education Level

| Dependent variable: Log inflation-adjusted income |                                |                                   |                                               |                                                  |
|---------------------------------------------------|--------------------------------|-----------------------------------|-----------------------------------------------|--------------------------------------------------|
| Variables                                         | OLS                            |                                   | IV                                            |                                                  |
|                                                   | (1)<br>High school<br>or below | (2)<br>Post-secondary<br>or above | (3)<br>High school<br>or below<br>(5-year MA) | (4)<br>Post-secondary<br>or above<br>(5-year MA) |
| Service length                                    | 0.003<br>(0.003)               | 0.002<br>(0.003)                  | 0.048<br>(0.038)                              | -0.002<br>(0.019)                                |
| Weak identification test                          |                                |                                   | 6.117                                         | 13.54                                            |
| Observations                                      | 13,959                         | 17,271                            | 13,959                                        | 17,271                                           |
| R-squared                                         | 0.280                          | 0.314                             | 0.165                                         | 0.317                                            |

Note: The instrument is the 5-year moving average (MA) of North Korean provocations prior to conscription. The full sample is split by education level. Service length is measured in months. All models controlled for education, age, the age at conscription, military spending for each cohort, class, and type of a respondent's job. Year and region fixed effects were also included. The standard errors are clustered by the survey years and the years when conscription started. See the first stage results in Table 13.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

There appears to be little change to the conscripts' level of human capital due to changes in the length of service. Existing literature finds that less educated men tend to benefit from conscription. While my findings do not contradict that of existing literature, I find that such a benefit is not primarily driven by positive human capital accumulation.

The lengths of conscription may also have different effects on people with different types of jobs. In Table 7, I examine if the service length may have different effects for males with office and non-office (manual) jobs. As shown in the table below, I split the sample by whether a respondent has an office job. I find no significant results in both the OLS and the IV models. It seems like the type of job that a male holds is irrelevant to the effect of conscription length on income.

Table 7: Results by Type of Jobs

| Dependent variable: Log inflation-adjusted income |                  |                  |                       |                           |
|---------------------------------------------------|------------------|------------------|-----------------------|---------------------------|
| Variables                                         | OLS              |                  | IV                    |                           |
|                                                   | (1)              | (2)              | (3)                   | (4)                       |
|                                                   | Office           | Non-office       | Office<br>(5-year MA) | Non-office<br>(5-year MA) |
| Service length                                    | 0.001<br>(0.002) | 0.003<br>(0.002) | 0.024<br>(0.028)      | -0.012<br>(0.040)         |
| Weak identification test                          |                  |                  | 13.76                 | 9.904                     |
| Observations                                      | 14,296           | 15,733           | 14,296                | 15,733                    |
| R-squared                                         | 0.461            | 0.333            | 0.408                 | 0.279                     |

Note: The instrument is the 5-year moving average (MA) of North Korean provocations prior to conscription. The full sample is split by whether a respondent holds an office job or not. Service length is measured in months. All models controlled for education, age, the age at conscription, military spending for each cohort, class, and type of a respondent's job. Year and region fixed effects were also included. The standard errors are clustered by the survey years and the years when conscription started. See the first stage results in Table 14. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

In all, a longer conscription length does not lead to a higher wage for people of different educational backgrounds or different types of jobs. That suggests there is little net human capital gain for conscripts. Conscription affects labor market outcomes primarily through the extensive margin.

## 5 Employment and Educational Attainment

Hjalmarsson and Lindquist (2019) find conscripts are more likely to commit crimes. That suggests conscription may have an effect on a conscript's chance

of employment. I study the effect of conscription length on an individual's chance of finding work here. Table 8 has the results of the analysis. See the first stage results in Table 15. The dependent variable is equal to 1 if a respondent is working and 0 otherwise. The rest of the specifications remain the same as before. In all columns, I find no significant results. Conscription lengths have no effect on the unemployment rate.

Table 8: Conscription Lengths and Employment Status

| Dependent variable: Respondent is employed |                  |                   |                   |                   |
|--------------------------------------------|------------------|-------------------|-------------------|-------------------|
| Variables                                  | OLS              | IV                |                   |                   |
|                                            | (1)<br>OLS       | (2)<br>6-year MA  | (3)<br>5-year MA  | (4)<br>4-year MA  |
| Service length                             | 0.000<br>(0.000) | -0.001<br>(0.001) | -0.001<br>(0.001) | -0.002<br>(0.001) |
| Weak identification test                   |                  | 12.33             | 11.44             | 5.070             |
| Observations                               | 31,230           | 31,226            | 31,230            | 31,230            |
| R-squared                                  | 0.931            | 0.930             | 0.930             | 0.928             |

Note: The dependent variable is equal to 1 if a respondent is working and 0 otherwise. The instruments are the 6-year, 5-year, and 4-year moving averages (MA) of North Korean provocations prior to conscription. The dependent variable is equal to 1 if a respondent is employed. All models controlled for education, age, age of conscription, and military spending for each cohort. Year and region fixed effects were also included. The standard errors are clustered by the survey years and the years when conscription started. See the first stage results in Table 15.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Taking breaks during one's education can lead to human capital depreciation. It is therefore possible that those who serve a longer term in the military may suffer more human capital depreciation. And more human capital depreciation can certainly in some cases lead to lower educational attainment. I study how conscription length can affect an individual's chance of obtaining an advanced degree in this section.

I examine the effect of service lengths on an individual's chance of obtaining post-secondary or post-graduate degrees and report the results in the first two columns of Table 9. The dependent variable in those columns is equal to 1 if a

respondent holds a post-secondary or post-graduate degree and 0 otherwise. Only the 6-year and 5-year moving averages of North Korean provocation events are used as instruments. See the first stage results in table 16. As shown in the table, both the OLS and the IV models find no significant results. That means shorter conscription length does not affect an individual's chance of obtaining a post-secondary degree. As [Toyryla \(2023\)](#) points out, most Korean males complete their military service 1 or 2 years after they start university. Korean males rarely complete their military service after university graduation. According to [Enthoven \(2013\)](#), even those who go to universities overseas complete their mandatory military service during their university years.

Table 9: Conscription Lengths and Educational Attainment

| Dependent variable: Respondent holds the following degrees |                                |                  |                      |                      |
|------------------------------------------------------------|--------------------------------|------------------|----------------------|----------------------|
| Variables                                                  | Post-secondary degree or above |                  | Post-graduate degree |                      |
|                                                            | (1)                            | (2)              | (3)                  | (4)                  |
|                                                            | 6-year MA                      | 5-year MA        | 6-year MA            | 5-year MA            |
| Service length                                             | -0.001<br>(0.020)              | 0.014<br>(0.021) | -0.031***<br>(0.010) | -0.030***<br>(0.009) |
| Weak identification test                                   | 13.25                          | 12.78            | 13.25                | 12.78                |
| Observations                                               | 30,026                         | 30,030           | 30,026               | 30,030               |
| R-squared                                                  | 0.415                          | 0.403            | 0.070                | 0.091                |

Note: The dependent variable in the first two columns is equal to 1 if a respondent holds a post-secondary or above and 0 otherwise. The dependent variable in the last two columns is equal to 1 if a respondent holds a post-graduate degree and 0 otherwise. The instruments are the 6-year and 5-year moving averages (MA) of North Korean provocations prior to conscription. Service length is measured in months. All models controlled for age, age of conscription, military spending for each cohort, class, and type of a respondent's job. Year and region fixed effects were also included. The standard errors are clustered by the survey years and the years when conscription started. See the first stage results in Table 16.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Although South Korean males' chance of going to college is not affected by the lengths of their military service, it is still possible that their chance of obtaining a post-graduate degree may be affected. Since most Korean men finish their CMS during university, a longer duration of conscription can lead

to more human capital depreciation on knowledge acquired from university courses. Such loss of human capital during university may seriously lower their grades, desire for a post-graduate degree, and chance of being admitted into a graduate school. I study whether longer conscription length affects people's chance of finishing graduate school and report the results in columns 3 and 4 of Table 9. I find that service length has a negative and significant effect on a conscript's chance of obtaining a post-graduate degree. An additional month of military service lowers an individual's chance of obtaining a post-graduate degree by about 3.1 percentage points. The unsurprising results confirm that longer breaks during one's education may lead to human capital depreciation.

Serving a longer term in the military does not hurt a conscript's chance of finishing college or chance of employment. However, a longer service in the military does lower his chance of obtaining a post-graduate degree. Ultimately, an individual has to do well in their undergraduate career to enter graduate school. The human capital depreciation may severely lower their GPA and prevent them from entering and finishing graduate school. In all, longer breaks only appear to affect an individual's chance of finishing graduate school because most people complete their military service during university.

## 6 Conclusion

There is a paucity of literature on the mechanism through which conscription affects labor market outcomes and educational attainment. By studying how conscription length affects those outcomes, I am able to control for the effects on the extensive margin and focus on the changes through the intensive margin due to changes in conscription length. Using North Korean provocation events as an instrumental variable and data from South Korea, I find the length of conscription does not have an effect on income, the chance of obtaining post-secondary degrees, or the employment status of conscripts. However, conscripts' chance of obtaining a post-graduate degree is lower when the conscription length is longer.

However, this paper is not without caveats. Since the South Korean

government always changes the duration of conscription by no more than 3 months, there may not be sufficient variation in the independent variable. Therefore, it is possible that there is not enough power to identify the effects of longer duration on labor market outcomes.

The results suggest that conscription affects labor market outcomes primarily through the extensive margin. Since longer breaks from work or study will cause human capital depreciation, a longer mandatory military service can certainly be costly and has few benefits to conscripts even during peacetime. However, the results also suggest that such costs hardly impact labor market outcomes and are primarily borne by those who seek to pursue post-graduate degrees. For countries that already have conscription and are suffering from a shortage of military personnel, the findings of this paper suggest that those countries can slightly increase the lengths of their compulsory military service without ruining conscripts' labor market prospects. Additionally, exceptions should be made for those pursuing post-graduate degrees.

In the Russo-Ukrainian War, both sides relied heavily on conscripts. On the other side of the world, the U.S. and Canadian armed forces are beginning to have recruitment shortages in 2023. While conscription may be on its way out, recruitment shortages and possible future high-intensity armed conflicts can force countries to reconsider their positions on conscription. The findings of this paper can help governments with their cost-benefit analysis of the adoption or abolishment of the institution.



## References

- Ari, B. P. (2019). Changes in military service period in the past (history of change) and the reason. <https://agibbyeongari.tistory.com/303>.
- Bauer, T. K., Bender, S., Paloyo, A. R., and Schmidt, C. M. (2012). Evaluating the labor-market effects of compulsory military service. *European Economic Review*, 56(4):814–829.
- Bingley, P., Lyk-Jensen, S. V., and Rosdahl, A. (2022). Military service and skill acquisition: Evidence from a draft lottery. *Journal of Human Capital*, 16(3):404–436.
- CSIS (2019). Database: North Korean Provocations - Beyond Parallel — beyondparallel.csis.org. <https://beyondparallel.csis.org/database-north-korean-provocations/>. [Accessed 17-09-2023].
- Enthoven, J. (2013). Korean students balance military service, academics — stanforddaily.com. <https://stanforddaily.com/2013/04/08/korean-students-balance-military-service-academics/>. [Accessed 21-09-2023].
- Hjalmarsson, R. and Lindquist, M. J. (2019). The causal effect of military conscription on crime. *The Economic Journal*, 129(622):2522–2562.
- Hofverberg, E. (2022). FALQs: The Conscription System of South Korea | In Custodia Legis — blogs.loc.gov. <https://blogs.loc.gov/law/2022/12/falqs-the-conscription-system-of-south-korea/>. [Accessed 26-02-2024].
- Huntington-Klein, N. (2021). Human capital versus signaling is empirically unresolvable. *Empirical Economics*, 60(5):2499–2531.
- KLI (2022). Klips — kli.re.kr. [https://www.kli.re.kr/klips\\_eng#secondPage](https://www.kli.re.kr/klips_eng#secondPage). [Accessed 17-09-2023].

- Lee, D. and Jung, Y. (2020). The effectiveness of military service experience on job experience. *Journal of Advances in Military Studies*, 3(3):91–112.
- Lee, M. (2022). [WHY] South Koreans and their apparent lack of concern over their northern neighbors. <https://koreajoongangdaily.joins.com/2022/08/27/national/northKorea/Korea-North-Korea-nuclear-weapons/20220827070006818.html>.
- Lim, H. (2018). Compulsory military service, civilian wages, and retirement decision: Evidence from korean individual-level data. *International Journal of Manpower*, 39(1):106–132.
- Puhani, P. A. and Sterrenberg, M. (2021). Effects of mandatory military service on wages and other socioeconomic outcomes.
- Savcic, R., Theodoropoulos, N., and Xefferis, D. (2023). Conscription and educational outcomes. *Journal of Population Economics*, pages 1–26.
- Toyryla, L. (2023). Military Service in Korea — 90daykorean.com. <https://www.90daykorean.com/military-service-in-korea/#:~:text=Typically%2C%20Korean%20men%20will%20first,they%20start%20their%20military%20service>. [Accessed 17-09-2023].
- Wallace, R. D. (2016). *North Korea and the Science of Provocation: Fifty Years of Conflict-Making*. McFarland.
- WB (2022). World Bank Open Data — data.worldbank.org. <https://data.worldbank.org/indicator/MS.MIL.XPND.GD.ZS?locations=KR>. [Accessed 17-09-2023].
- Yoo, S. and Kim, S. (2017). The pattern of north korea’s local military provocations. *The Korean Journal of International Studies*, 15(1):71–84.

## First Stage Results

Table 10: Conscription Lengths and Income

| Variables                         | Instruments          |                      |                      |
|-----------------------------------|----------------------|----------------------|----------------------|
|                                   | (1)<br>No. of NK     | (2)<br>provocation   | (3)<br>events        |
| 6-year moving average             | 0.309***<br>(0.089)  |                      |                      |
| 5-year moving average             |                      | 0.257***<br>(0.076)  |                      |
| 4-year moving average             |                      |                      | 0.165**<br>(0.074)   |
| Age                               | 0.248***<br>(0.029)  | 0.264***<br>(0.024)  | 0.291***<br>(0.023)  |
| Log military spending as % of GDP | -0.525<br>(0.840)    | -0.838<br>(0.703)    | -1.515**<br>(0.672)  |
| Level of education (schooling)    | 0.213*<br>(0.116)    | 0.210*<br>(0.117)    | 0.205*<br>(0.117)    |
| Work status (wage/ non-wage)      | 0.088<br>(0.197)     | 0.091<br>(0.196)     | 0.085<br>(0.196)     |
| Age conscripted                   | -0.424***<br>(0.073) | -0.437***<br>(0.073) | -0.457***<br>(0.073) |
| F-statistic                       | 12.05                | 11.29                | 4.964                |
| Observations                      | 31,226               | 31,230               | 31,230               |

Note: region, year, and all job fixed effects are omitted.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 11: Mean Income and Duration of Conscription

| Variables                         | Instruments                         |                     |                     |
|-----------------------------------|-------------------------------------|---------------------|---------------------|
|                                   | (1)<br>No. of NK provocation events | (2)                 | (3)                 |
| 6-year moving average             | 0.525***<br>(0.084)                 |                     |                     |
| 5-year moving average             |                                     | 0.463***<br>(0.077) |                     |
| 4-year moving average             |                                     |                     | 0.379***<br>(0.075) |
| Age                               | 0.133***<br>(0.021)                 | 0.140***<br>(0.021) | 0.150***<br>(0.022) |
| Log military spending as % of GDP | 2.655***<br>(0.559)                 | 2.554***<br>(0.521) | 2.287***<br>(0.551) |
| Age conscripted                   | -0.142**<br>(0.057)                 | -0.144**<br>(0.057) | -0.142**<br>(0.057) |
| F-statistic                       | 39.47                               | 36.61               | 25.68               |
| Observations                      | 4,297                               | 4,300               | 4,300               |

Note: education-level fixed effects are omitted.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 12: Parsimonious Models

| Variables                         | Instruments                         |                      |                      |
|-----------------------------------|-------------------------------------|----------------------|----------------------|
|                                   | (1)<br>No. of NK provocation events | (2)                  | (3)                  |
| 6-year moving average             | 0.309***<br>(0.078)                 |                      |                      |
| 5-year moving average             |                                     | 0.239***<br>(0.064)  |                      |
| 4-year moving average             |                                     |                      | 0.161***<br>(0.059)  |
| Age                               | 0.311***<br>(0.029)                 | 0.332***<br>(0.024)  | 0.356***<br>(0.023)  |
| Log military spending as % of GDP | -3.579***<br>(0.812)                | -4.025***<br>(0.683) | -4.619***<br>(0.657) |
| Level of education (schooling)    | 0.581***<br>(0.151)                 | 0.577***<br>(0.151)  | 0.573***<br>(0.151)  |
| Work status (wage/ non-wage)      | 1.028<br>(0.693)                    | 1.032<br>(0.683)     | 1.028<br>(0.681)     |
| Age conscripted                   | -0.701***<br>(0.092)                | -0.719***<br>(0.091) | -0.739***<br>(0.091) |
| F-statistic                       | 15.59                               | 13.95                | 7.547                |
| Observations                      | 32,729                              | 32,733               | 32,734               |

Note: region, year, industry, and job fixed effects are omitted.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 13: Results by Education Level

|                                   | High school<br>or below | Post-secondary<br>or above |
|-----------------------------------|-------------------------|----------------------------|
|                                   | (1)                     | (2)                        |
| Variables                         | No. of NK               | provocation events         |
| 5-year moving average             | 0.220**<br>(0.0890)     | 0.329***<br>(0.0891)       |
| Log military spending as % of GDP | -1.900*<br>(1.142)      | 0.201<br>(1.007)           |
| Level of education (schooling)    | 1.691***<br>(0.457)     | -0.111<br>(0.224)          |
| Work status (wage/ non-wage)      | 0.249<br>(0.329)        | -0.217<br>(0.220)          |
| Age conscripted                   | -0.441***<br>(0.107)    | -0.377***<br>(0.0972)      |
| Age                               | 0.331***<br>(0.0392)    | 0.222***<br>(0.0364)       |
| F-statistic                       | 6.117                   | 13.54                      |
| Observations                      | 13,959                  | 17,271                     |

Note: region, year, and all job fixed effects are omitted.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

Table 14: Results by Type of Jobs

| Variables                         | Office                       | Non-office           |
|-----------------------------------|------------------------------|----------------------|
|                                   | (1)                          | (2)                  |
|                                   | No. of NK provocation events |                      |
| 5-year moving average             | 0.315***<br>(0.085)          | 0.238***<br>(0.076)  |
| Log military spending as % of GDP | 0.877<br>(0.864)             | -1.991**<br>(0.825)  |
| Level of education (schooling)    | -0.121<br>(0.126)            | 0.567***<br>(0.177)  |
| Work status (wage/ non-wage)      | -0.028<br>(0.379)            | 0.557*<br>(0.303)    |
| Age conscripted                   | -0.383***<br>(0.094)         | -0.450***<br>(0.099) |
| Age                               | 0.194***<br>(0.029)          | 0.315***<br>(0.033)  |
| F-statistic                       | 13.76                        | 9.904                |
| Observations                      | 14,296                       | 15,733               |

Note: region, year, and all job fixed effects are omitted.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1

Table 15: Conscription Lengths and Employment Status

| Variables                         | Instruments          |                      |                      |
|-----------------------------------|----------------------|----------------------|----------------------|
|                                   | (1)<br>No. of NK     | (2)<br>provocation   | (3)<br>events        |
| 6-year moving average             | 0.311***<br>(0.089)  |                      |                      |
| 5-year moving average             |                      | 0.259***<br>(0.077)  |                      |
| 4-year moving average             |                      |                      | 0.167**<br>(0.074)   |
| Log military spending as % of GDP | -0.488<br>(0.842)    | -0.808<br>(0.703)    | -1.488**<br>(0.673)  |
| Level of education (schooling)    | 0.214*<br>(0.116)    | 0.211*<br>(0.117)    | 0.207*<br>(0.116)    |
| Age conscripted                   | -0.422***<br>(0.073) | -0.435***<br>(0.073) | -0.456***<br>(0.073) |
| Age                               | 0.247***<br>(0.029)  | 0.263***<br>(0.024)  | 0.291***<br>(0.023)  |
| F-statistic                       | 12.33                | 11.44                | 5.070                |
| Observations                      | 31,226               | 31,230               | 31,230               |

Note: region, year, and all job fixed effects are omitted. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1



Table 16: Conscription Lengths and Educational Attainment

| Variables                         | Post-secondary<br>degree or above |                      | Post-graduate<br>degree |                      |
|-----------------------------------|-----------------------------------|----------------------|-------------------------|----------------------|
|                                   | (1)                               | (2)                  | (3)                     | (4)                  |
|                                   | No. of NK provocations events     |                      |                         |                      |
| 6-year moving average             | 0.304***<br>(0.088)               |                      | 0.304***<br>(0.088)     |                      |
| 5-year moving average             |                                   | 0.254***<br>(0.077)  |                         | 0.254***<br>(0.077)  |
| Log military spending as % of GDP | -0.428<br>(0.858)                 | -0.736<br>(0.720)    | -0.428<br>(0.858)       | -0.736<br>(0.720)    |
| Work status (wage/ non-wage)      | 0.095<br>(0.196)                  | 0.098<br>(0.195)     | 0.095<br>(0.196)        | 0.098<br>(0.195)     |
| Age conscripted                   | -0.400***<br>(0.073)              | -0.413***<br>(0.073) | -0.400***<br>(0.073)    | -0.413***<br>(0.073) |
| Age                               | 0.237***<br>(0.029)               | 0.253***<br>(0.024)  | 0.237***<br>(0.029)     | 0.253***<br>(0.024)  |
| F-statistic                       | 13.25                             | 12.78                | 13.25                   | 12.78                |
| Observations                      | 31,228                            | 31,232               | 31,228                  | 31,232               |

Note: region, year, and all job fixed effects are omitted. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1